MIL-A-11221A (Ord)
22 February 1962
SUPERSEDING
MIL-P-11221 (Ord)
20 June 1951
(See 6.2)

MILITARY SPECIFICATION

ARMING PLUG, MINE, ANTITANK M4 (PRACTICE) AND M4A1

1. SCOPE

1.1 This specification covers the parts and assembly for Arming Plug, Mine, Antitank, M4 and M4A1 (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons, General Specification for.
MIL-P-10025 - Packing and Marking for Domestic Shipment of Inert Ammunition Components; General Specification for.
MIL-I-45208 - Inspection Requirements, General Specification for.
MIL-I-45607 - Inspection Equipment, Supply and Maintenance of.

FSC 1345

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109 - Inspection Terms and Definitions.

DRAWINGS

ORDNANCE CORPS

7548160 - Arming Plug, Mine, Anti-Tank, Practice, M4, Assembly.
8860866 - Arming Plug, Mine, Anti-Tank, M4A1, Assembly.

PUBLICATIONS

ORDNANCE CORPS

ORD-M608-11 - Procedures and Tables for Continuous Sampling by Attributes.
IEL-7299796 - Index of Inspection Equipment Lists.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.-Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Arming plug assemblies.-The arming plug assemblies shall comply with all requirements specified on Drawing (dwg.) 7548160 or 8860866, as applicable, and with all requirements specified in applicable specifications.

3.3 Internal air pressure.-The arming plug assemblies shall withstand an internal air pressure of 10 pounds per square inch gage (p.s.i.g.) for not less than two seconds with no evidence of leakage when tested as specified in 4.4.1.

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3.4 Strength of welds.-The two spot welds, or the alternative single spot weld, as applicable, shall pull a button of a diameter not less than 75 percent of the diameter of the weld specified on the applicable drawing when tested as specified in 4.4.4.

3.5 Load test of pins.-The shutter guide pin and shutter bearing pin subassemblies shall be riveted to withstand a force of not less than 175 pounds when tested as specified in 4.4.3.

3.6 Torque of setting knob.-The arming plugs shall arm when a torque of not less than two nor more than eight inch-pounds is applied to the setting knob when tested as specified in 4.4.2.

3.7 Workmanship.-All parts shall be free of chips, weld splatter, slag, scale, dirt, grease, rust, and other foreign material. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-109 in order to define terms used herein. The provisions of Specification MIL-A-2550 shall apply.

4.1.1 Contractor quality assurance system.-The contractor shall provide and maintain an effective quality assurance system in compliance with the requirements of Specification MIL-I-45208.

4.1.2 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a

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certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

b. Number of units of product inspected.

c. Results obtained for all inspection performed.

d. Drawing, specification number and date, together with an identification and date of changes.

e. Certificates of analysis on all material procured directly by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

f. Number of items in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification.-Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings, and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be in accordance with Specification MIL-I-45208 and will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Initial production inspection

4.2.1 Submission.-The contractor shall submit an initial production sample consisting of twenty-five (25) consecutively produced assemblies and twenty-five (25) complete sets of parts, which have been produced by the production process which the contractor will use in fulfilling the contract, to a Government approved facility designated by the contracting officer. All parts and materials shall be obtained from the same source of supply as will be used in regular production. The initial production sample shall be accompanied by certificates

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of analyses for all materials. An initial production sample shall be submitted prior to initiation of production and whenever a change occurs in manufacturing process, material used, drawing, specification, or source of supply such as to significantly affect product uniformity as determined by the Government.

4.2.2 Inspections to be performed.-Each assembly and component shall be inspected by the Government for all the inspections specified in 4.3.2 and 4.4 of this specification and for all requirements of the applicable drawings.

4.2.3 Rejection.-If any assembly or component fails to comply with the applicable requirements, the initial production sample, shall be rejected and the contractor shall take the necessary corrective action and submit a new initial production sample, or portion thereof, as specified by the contracting officer. The contractor shall continue to submit new or supplemental quantities as necessary until such time as the sample quantity successfully passes the inspection specified or until otherwise directed by the procuring activity.

4.3 Inspection provisions.

4.3.1 Lot formation.-A lot shall consist of arming plug assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit, as determined by the Government shall not necessitate changing the lot number. Each lot shall contain arming plug assemblies of one type only.

4.3.2 Examination.-Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-105 except that inspection for critical defects, when listed, shall be 100 percent. Continuous sampling plans in accordance with Handbook ORD-M608-11 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed using an AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Workmanship.-The following applies to all components not hereinafter specifically listed in 4.3.2.2 through 4.3.2.13 (see 3.7 and applicable drawings).

Categories	Defects	Method of Inspection	Code No 0.
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Operation missing or improperly performed.....	Visual	01001
102.	Burr or chip on or at intersection of sliding or other functional surfaces.....	Visual	01002
103.	Finish of sliding or other functional surface improper...	Visual	01003
104.	Component missing or improperly assembled.....	Visual	01004
105.	Protective finish inadequate..	Visual	01005
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship other than that covered under major category above (see 3.7).....	Visual	01006

4.3.2.2 Pin, shutter guide (see dwg. 7548166 covering a detail of dwg. 7548160 (M4) and 8860866 (M4A1)).

Categories	Defects	Method of Inspection	Code No 02
Critical:	None defined.		
Major:	None defined.		
Minor:	AQL 1.50 percent		
201.	Smaller outside diameter.....	Gage	02001
202.	Length to flange.....	Gage	02002
203.	Diameter of flange, minimum (min.).....	Gage	02003

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4.3.2.3 Cup (see dwg. 7548162 covering a detail of dwg. 7548160 (M4) and 8860866 (M4A1))

Categories	Defects	Method of Inspection	Code No 03
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Outside diameter below flange..	Gage	03001
102.	Outside diameter at bottom....	Gage	03002
103.	Distance between setting knob seat and flange.....	Gage	03003
104.	Total height.....	Gage	03004
105.	Thickness of strip, minimum (min.).....	Gage	03005
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.7).....	Visual	03006

4.3.2.4 Knob, setting (see dwg. 7548164 covering a detail of dwg. 7548160 (M4) and 8860866 (M4A1))

Categories	Defects	Method of Inspection	Code No 04
Critical:	None defined.		
Major:	AQL 0.40 percent		
101.	Outside diameter below flange, min.....	Gage	04001
102.	Thickness of flange, min.....	Gage	04002
103.	Depth to bottom.....	Gage	04003
104.	Diameter of flange, min.....	Gage	04004
Minor:	AQL 0.40 percent		
201.	Evidence of poor workmanship (see 3.7).....	Visual	04005

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4.3.2.5 Spring retainer (see dwg. 8860868 covering a detail of dwg. 8860866) (M4A1)

Categories	Defects	Method of Inspection	Code No 05
Critical: None defined.			
Major: None defined.			
Minor:	AQL 2.5 percent		
201.	Outside diameter.....	Gage	05001
202.	Thickness through slit wall, min.....	Gage	05002
203.	Total height.....	Gage	05003
204.	Crack in radius.....	Visual	05004
205.	Evidence of poor workmanship (see 3.7).....	Visual	05005

4.3.2.6 Shutter (see dwg. 7548168 covering a detail of dwg. 7548160 (M4) and 8860866 (M4A1)

Categories	Defects	Method of Inspection	Code No 06
Critical: None defined.			
Major: None defined.			
Minor:	AQL 2.5 percent		
201.	Width of slot, max.....	Gage	06001
202.	Length of slot.....	Gage	06002
203.	Distance between slot and hole.....	Gage	06003
204.	Diameter of hole, max.....	Gage	06004
205.	Location of hole.....	Gage	06005

4.3.2.7 Pin, shutter bearing (see dwg. 7548165 covering a detail of dwg. 7548160 (M4) and 8860866 (M4A1)

Categories	Defects	Method of Inspection	Code No 07
Critical: None defined.			

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Major: None defined.

Minor:	AQL 1.0 percent		
201.	Length to flange.....	Gage	07001
202.	Diameter below flange, min....	Gage	07002

4.3.2.8 Spring (see dwg. 8860869 covering a detail of dwg. 8860866). (M4A1)

Categories	Defects	Method of Inspection	Code No 08
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Load at .85 inches, min.....	Gage	08001
102.	Load at .50 inches, min.....	Gage	08002
103.	Outside diameter of larger end, min.....	Gage	08003
Minor:	AQL 1.5 percent		
201.	Outside diameter of smaller end, min.....	Gage	08004
202.	Wire with cuts or deep nicks.....	Visual	08005
203.	Protective coating damaged (base metal exposed).....	Visual	08006

4.3.2.9 Fuze retainer (see dwg. 8860870 covering a detail of dwg. 8860866). (M4A1)

Categories	Defects	Method of Inspection	Code No 09
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Inside diameter.....	Gage	09001
102.	Total height.....	Gage	09002
Minor:	AQL 1.5 percent		
201.	Dimple missing.....	Visual	09003
202.	Crack in radius.....	Visual	09004
203.	Evidence of poor workmanship (see 3.7).....	Visual	09005

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4.3.2.10 Body (see dwg. 7548161 covering a detail of dwg. 7548160 and 8860866) (M4 and M4A1, resp.)

Categories	Defects	Method of Inspection	Code No 10
Critical:	None defined.		
Major:	AQL 1.0 percent		
101.	Major diameter of thread.....	Gage	10001
102.	Pitch diameter of thread.....	Gage	10002
103.	Length to gasket seat above thread.....	Gage	10003
104.	Perpendicularity of gasket seat behind flange with thread.....	Gage	10004
105.	Diameter of cup hole, max.....	Gage	10005
106.	Inside diameter, max.....	Gage	10006
107.	Diameter between threads and gasket seat.....	Gage	10007
108.	Diameter of flange, min.....	Gage	10008
109.	Marking improperly positioned..	Visual	10009
Minor:	AQL 1.5 percent		
201.	Thickness through bottom, min.....	Gage	10010
202.	Width of flange lug, min.....	Gage	10011
203.	Finish of cup face improper...	Visual	10012
204.	Evidence of poor workmanship (see 3.7).....	Visual	10013

4.3.2.11 Assembly (see dwg. 7548160) (M4)

Categories	Defects	Method of Inspection	Code No 11
Critical:	None defined.		
Major:	AQL 0.65 percent		
101.	Torque of setting knob (see 3.6).....	Test	11001
102.	Shutter does not clear a plug gage 1.062 inches in diameter when inserted into the cup (arrowhead facing index line between the words "DANGER" and "ARMED".....	Gage	11002

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103.	Distance between body flange and shutter face, max.....	Gage	11003
104.	Gap between body and setting knob, min.....	Gage	11004
105.	Shutter or setting knob improperly assembled (with red arrowhead facing the word "SAFE", hold assembly with notched lug facing up and red arrowhead toward inspector. The word "ARMED" must be to the left of the arrowhead, shutter slot under the arrowhead, with the shutter guide pin engaged in the shutter slot and riveted to the cup and the shutter bearing pin riveted to the body.....	Visual	11005
106.	Shutter not in armed position when red arrowhead on setting knob is set to index line between the words "ARMED" and "DANGER"....	Visual	11006
107.	Gasket missing.....	Visual	11007
Minor:	AQL 2.5 percent		
201.	Distance across flange lugs...	Gage	11008
202.	Thread damaged.....	Visual	11009
203.	Offset missing from lugs.....	Visual	11010
204.	Coating damaged (base metal exposed).....	Visual	11011
205.	Gasket defective (cut, torn, etc.).....	Visual	11012
206.	Evidence of poor workmanship (see 3.7).....	Visual	11013
207.	Marking misleading or unidentifiable, including color code.....	Visual	11014

4.3.2.12 Assembly (see dwg. 8860866).(M4A1)

Categories	Defects	Method of Inspection	Code No 12
Critical:			
1.	Shutter does not clear a plug gage 1.062 inches in diameter when inserted into the cup		

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	(arrowhead facing index line between words "DANGER" and "SAFE").....	Gage	12001
2.	Shutter or setting knob improperly assembled (with red arrowhead facing the word "SAFE", hold assembly with notched lug facing up and spring facing away from inspector. The word, "ARMED" must be to left of arrowhead, shutter slot under the arrowhead, with the shutter guide pin engaged in the shutter slot and riveted to the cup and the shutter bearing pin riveted to the body.....	Visual	12002
Major:	AQL 0.65 percent		
101.	Torque of setting knob (see 3.6).....	Test	12003
102.	Distance between body flange and shutter face, max..	Gage	12004
103.	Gap between body and setting knob, min.....	Gage	12005
104.	Shutter not in armed position when red arrowhead on setting knob is set to index line between words "ARMED" and "DANGER".....	Visual	12006
105.	Fuze retainer or spring retainer crimp inadequate (spring wire not retained by crimp or bent strip).....	Visual	12007
106.	Gasket missing.....	Visual	12008
107.	Fuze retainer missing.....	Visual	12009
Minor:	AQL 2.5 percent		
201.	Distance across flange lugs, min.....	Gage	12010
202.	Thread damaged.....	Visual	12011
203.	Offset missing from lugs.....	Visual	12012
204.	Coating damaged (base metal exposed).....	Visual	12013
205.	Gasket defective (cut, torn, etc.).....	Visual	12014

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206.	Evidence of poor workmanship (see 3.7).....	Visual	12015
207.	Marking misleading or unidentifiable, including color code.....	Visual	12016

4.3.2.13 Carton, sealed (see section 5)

Categories	Defects	Method of Inspection	Code No 13
Critical:	None defined.		
Major:	AQL 0.25 percent		
101.	Assembly damaged so that contents are exposed or liable to become exposed.....	Visual	13001
102.	Sealing strip incomplete, badly wrinkled, or separation at any point around carton exceeds 1/2 inch.....	Visual	13002
Minor:	AQL 1.0 percent		
201.	Contents move when carton is shaken.....	Manual	13003
202.	Marking misleading or unidentifiable.....	Visual	13004

4.3.3 Testing

4.3.3.1 Internal air pressure (see 3.3) Major defect.- The sampling plan for this test shall be in accordance with Standard MIL-STD-105 using code letter L and an AQL of 0.25 percent. The test shall be performed as specified in 4.4.1 using equipment in accordance with 4.3.4. Any arming plug assembly which fails to comply with the specified requirement shall be classed defective. (11015)M4 (12017)M4A1

4.3.3.2 Load test of pins (see 3.5) Major defect.-The sampling plan for this test shall be in accordance with Standard MIL-STD-105 using code letter K and an AQL of 0.40 percent. The test shall be performed as specified in 4.4.3 using equipment in accordance with 4.3.4. Any pin which fails to comply with the specified requirements shall cause the arming plug assembly to be classed defective. (11016)M4 (12018)M4A1

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4.3.3.3 Strength of welds (see 3.4) Major defect.-The sampling plan for these tests shall be in accordance with Standard MIL-STD-105 using code letter H and an AQL of 2.5 percent. The tests shall be performed as specified in 4.4.4. using equipment in accordance with 4.3.4. Any assembly which fails to comply with the specified requirements shall be classed defective. (11017)M4 (12019)M4A1

4.3.4 Inspection equipment.-Index of Inspection Equipment Lists Number 7299796 identifies available approved inspection equipment designs. The contractor may, however, design alternate inspection equipment in order to facilitate his operation. Designs of any acceptance inspection equipment prepared by the contractor to support his approved written inspection plan (see 4.1.1) shall be approved by the Government prior to fabrication or procuring of such equipment (see 6.4). All acceptance inspection equipment shall be certified by the Government as complying with the approved design prior to its use. Supply and maintenance of inspection equipment shall be in accordance with Specification MIL-I-45607.

4.4 Test methods and procedures.

4.4.1 Internal air pressure.-The arming plug assembly shall be assembled in a specified testing device and subject to a minimum air pressure of 10 p.s.i.g. for not less than 2 seconds to determine compliance with 3.3.

4.4.2 Torque of setting knob.-The arming plug assembly shall be placed in a specified device and the setting knob rotated from the safe to the armed position to determine compliance with the torque requirement specified in 3.6.

4.4.3 Load test of pins.-The arming plug assembly, less setting knob, gasket and fuze retainer assembly (M4A1), shall be assembled to a specified device and a load not less than 175 pounds applied in turn, to the riveted end of the shutter guide pin and shutter bearing pin to determine compliance with 3.5.

4.4.4 Strength of welds.-The arming plug assemblies, less gasket and fuze retainer (M4A1), shall be tested to destruction by any method satisfactory to the Government inspector to determine compliance with 3.4.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level C.-See 5.2.1 for requirements.

5.2 Packing

5.2.1 Level C.-The arming plugs shall be packed in accordance with Military Specification MIL-P-10025. Springs (M4A1) may be compressed and restrained with tape or other easily removed fastening prior to packing.

5.3 Marking.-Marking shall be in accordance with Military Specification MIL-P-10025.

5.4 Data cards.-Data card information shall be as specified in Military Specification MIL-A-2550.

6. NOTES

6.1 Ordering data.-Procurement documents should specify the title, number, and date of this specification.

6.2 Previous title.-The previous title of this specification was "Plug, Arming, AT, Mine, M4 and M4B1".

6.3 Intended use.-The M4 and M4A1 arming plugs are used with various practice and HE mines, respectively.

6.4 Submission of contractor-developed equipment designs.-Documentation of contractor-developed inspection equipment designs shall be submitted in duplicate and identified with the associated examination or test required by the specification, to Picatinny Arsenal, Dover, New Jersey, Attention: ORDBB-ND, for approval. This documentation shall include special equipment designs and a complete description of the use and application of all standard devices and set-ups as well as special equipment designs. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

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6.5 Inspection code numbers.-The two and five digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

Notice.-When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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